SE(3)-WAM: What Should a Robot Remember and Predict for Dynamic Manipulation?

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Abstract

Dynamic manipulation requires a robot to coordinate contact with a scene that continues to change between observation and execution. This raises a question beyond predicting more of the scene: which historical and future information actually improves the action? We study history encoding, future representation, and world–action coupling in a common dynamic-manipulation framework. Our guiding hypothesis is that prediction should preserve target motion and task-relevant context at the granularity required by interaction. A completed fifteen-condition future-supervision study on fourteen simulated tasks shows why this hypothesis requires controlled testing. On 1,260 paired evaluation scenes, the highest point estimates are 64.13% and 64.05%, compared with 62.78% without future supervision, with no clear overall advantage. Dense 2-D displacement supervision reaches 59.21% and is worse under the reported multiplicity correction. These results use privileged current object state and motion history, one training seed, and an auxiliary prediction path. History and combination studies further separate information access from representation choice. We specify the remaining backbone, online information-flow, pretraining, and external evaluations that will determine the final SE(3)-WAM instance. The present evidence supports conditional design decisions, while leaving the value of deployable online foresight to those comparisons.

1. Introduction

A moving object can leave its observed position before a robot reaches it. The same problem arises when a receptacle moves during release or an obstacle changes the available approach corridor. Dynamic manipulation therefore requires spatial accuracy at the right time. Faster inference and more responsive execution reduce observation age [31], while temporal observations help distinguish motions that look similar in a single image [8, 21]. Together, these demands motivate a practical question: *what should a robot remember and predict so that its action remains useful when interaction occurs?*

The answer depends on the role of information in the task. Target translation and rotation determine where and how to make contact. A moving occluder or obstacle matters through its effect on visibility, reachability, or collision. Static geometry can remain essential even though it has no motion to predict. This suggests organizing scene evolution by interaction needs: preserve fine target information when contact requires it, and represent surrounding changes at sufficient resolution to support the decision. Both a fixed target/context decomposition and a shared representation with task-dependent readout can implement this idea. Their relative value is an empirical question.

World–action models provide a framework for learning scene evolution alongside robot actions [34, 35]. However, a future objective changes several things at once: the supervised variables, the gradients shaping shared features, and sometimes the computation used before acting. The action may benefit from auxiliary supervision even when no forecast is generated at deployment [37]. Conversely, a geometrically accurate forecast may add no useful information to an already strong state-conditioned policy. Prediction error, action improvement, and online forecast use must therefore be measured separately.

We organize the SE(3)-WAM Study around three connected questions. **Q1** asks how a finite history should encode motion. **Q2** asks which future variables, regions, and resolutions support action. **Q3** asks how world and action computation should share parameters and exchange information. The study first establishes a common reference, compares history and future choices, and then selects a backbone and its information flow. Targeted cross-comparisons revisit earlier choices before the complete design is used for robot-data pretraining. This ordering exposes interactions between design decisions without requiring an exhaustive Cartesian product.

The completed evidence already constrains the story. Fifteen future-supervision conditions have been trained and evaluated on a 1,260-scene complement of the development

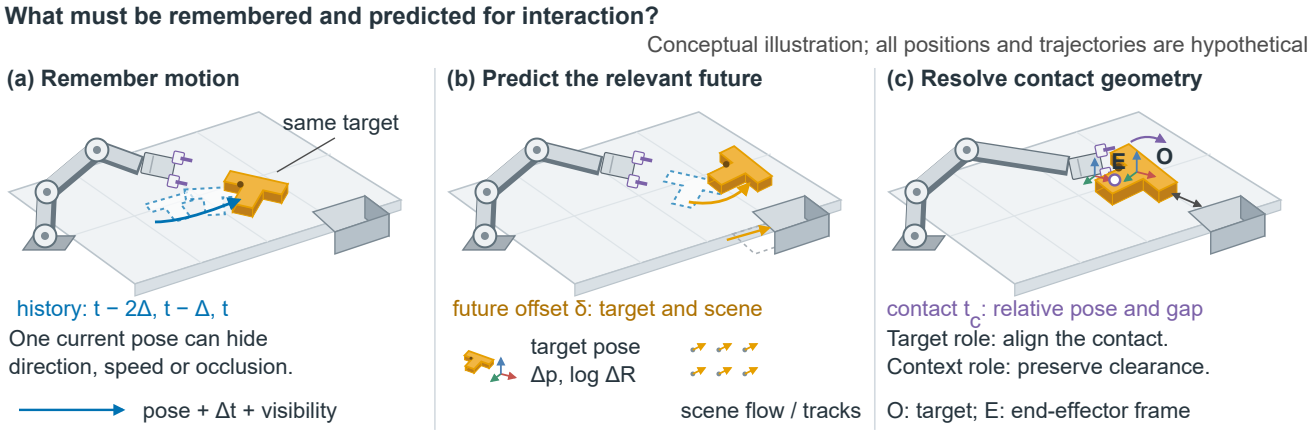


Figure 1. What information matters at interaction time? (a) History disambiguates motion hidden by a single target pose. (b) Future target pose and scene motion describe complementary physical quantities. (c) Contact requires end-effector/target alignment and clearance from context. The same asymmetric target and robot connect the panels. All trajectories are conceptual illustrations of the research question.

set. Sparse scene or target trajectories give the highest point estimates, but neither clearly exceeds the no-future reference overall. Dense 2-D future displacement decreases success under the same comparison. A subsequent twelve-condition combination screen selects a candidate with 66.59% success on that complement, while its exact matched controls remain incomplete. An adapted DreamVLA reference reaches 79.60% under its own recipe. These observations motivate investigating how information is encoded and used, rather than treating the addition of future targets as an established benefit.

Our contributions at this stage are:

- A dynamic-manipulation study protocol that separates task outcomes, available information, training supervision, and execution conditions.
- A completed comparison of fifteen future-supervision conditions, with paired results and adverse findings, complemented by history and combination studies that identify the next controlled contrasts.
- A unified design and evaluation framework for selecting SE(3)-WAM through history, future-representation, backbone, and information-flow comparisons, followed by a specified pretraining and transfer study.

The supplement provides complete configurations, results, and remaining protocols.

2. Related Work

Dynamic manipulation and physical time. Motion-aware grasping connects prediction to reachability and interception [1]. DOMINO/PUMA couples a dynamic benchmark with historical flow and object-future supervision [8]; DynamicManip expands static demonstrations into dynamic data [15]. DynamicVLA emphasizes continuous inference and action streaming [31], while DynamicWAM conditions world–action learning on motion [21]. Temporal feature prediction [40], AHEAD's predictive latents [25], and ForeTime-VLA's future-token and event-time supervision [22] offer different ways to anticipate interaction. RTC addresses continuity under delayed chunk generation [2]; Chronos Policy explicitly represents observation and execution times [27]. These works motivate evaluating prediction together with action age, rather than identifying a simulator control interval with real-time operation.

History and object-centered information. TraceVLA encodes temporal traces as visual prompts [42]; HiF-VLA organizes past and future dynamics through motion representations [16]. PACT-WAM allocates fewer spatial tokens to older observations [19], and Long-WAM studies long context together with causal video pretraining [11]. Thus a history encoder's value depends on both temporal support and the training that makes that information usable. DOCIR separates manipulated objects, obstacles, and the robot [7]; OA-WAM makes objects addressable within world–action modeling [20]. Our Q1 comparisons distinguish current localization, privileged motion, and visual history before comparing encodings. Role assignment is an experimental factor, not a presumed optimum.

Which future should a policy predict? SE3-Nets models scene motion through rigid transforms [3]; SE3-Pose-Nets uses structured pose dynamics for control [4]. DSR-Net accumulates a volumetric object representation over interactions [33], while PointWorld and PhysMani develop 3-D world representations for manipulation [10, 38]. ATM learns point trajectories from video [29]; Im2Flow2Act uses generated object flow as a policy interface [32]. AffordTrajDP

propagates contact affordances with object pose [30], and FlowWAM uses flow in policy and world-model modes [5]. These interfaces differ in whether they describe desired motion, action-conditioned consequences, or externally moving objects. DreamVLA and DreamWAM add complementary visual, geometric, and semantic targets [36, 39]; DC-WAM instead emphasizes dynamics within RGB supervision [12]. DELE-w0.5 uses compact future latents [14]. Q2 places related target families under one action path to measure their conditional training value; its pose output does not imply a new rigid-motion parameterization.

World–action coupling and experimental attribution. VPP conditions action learning on predictive video features [9], whereas DreamZero jointly models world and action dynamics [34]. Fast-WAM separates training benefits from test-time imagination [37]. Two distinct OpenWAM projects provide relevant design studies: Wang et al. investigate modular pretraining and release the Alpha foundation used by our adapted baseline [28]; Yu et al. compare composable world–action programs in a common framework [35]. Neither is the pretrained source of our freshly initialized reference D backbone. Q3 separates shared-parameter learning, cross-stream access, and generation order, then revisits representation choices. ACT, Diffusion Policy, and $\pi_{0.5}$ supply complementary system references [6, 23, 41]; DROID supplies the planned real-robot pretraining data [13]. Native system comparisons and matched design contrasts answer different questions and are reported separately.

3. Problem and Evaluation Protocol

Interaction in physical time. Let $\mathcal{H}_t = \{(o_i, s_i, \tau_i)\}_{i \le t}$ denote the available finite history of observations, robot state, and timestamps. A policy produces an action chunk $A_t = (a_{t,0}, \dots, a_{t,H_a-1})$. If the latest image was captured at τ_t^s and the first command is applied after delay ℓ_t, its application times are

$$\tau_{t,j}^{\text{apply}} = \tau_t^s + \ell_t + j\Delta_a. \tag{1}$$

The relevant future is defined relative to these physical times and the task's contact event. Forecast horizons, action intervals, and computation time are different quantities. In the completed representation screen, labels use eight future samples from 50 to 400 ms, while evaluation uses synchronous policy queries and executes one action per replan. Real computation delay is not injected into the physical evolution by those synchronous results.

Tasks and data. The reported experiments use the migrated DB0 v0.6 data and the fourteen-task evaluation suite. The suite covers moving-target grasping, orientation-sensitive grasping, physical occlusion, moving-receptacle placement, phase-constrained passage, free rolling/sliding motion, and post-impact adaptation. The simulation infrastructure uses MuJoCo and robosuite [26, 43]. Task success is determined by each native evaluator's physical outcome; all fourteen tasks receive equal weight. The common demonstration pool contains 11,200 trajectories, 800 per task. P0 motion-only prediction and later fresh-data production are separate from this learning/evaluation record. Appendix A describes the task and split identities used here.

Information tracks. The future-supervision screen uses a current fixed-camera RGB image, exact depth/camera geometry, an eight-sample robot-state history, and privileged current target pose plus seven intervals of target kinematics. We call this the *object-motion information track*. It isolates future objectives when substantial target motion information is already available. The current-pose history study removes privileged interval kinematics and asks whether visual history supplies useful additional information. Both tracks still use privileged current target pose. A deployable visual system must estimate such quantities from allowed sensors and is evaluated separately. Current-only RGB with robot-state history is denoted H0V; it retains memory.

Evaluation partitions. The fixed140 development set contains ten scenes per task. The disjoint 1,260-scene complement contains ninety per task, giving 1,400 scenes together. The complement is independent of fixed140 membership but belongs to the same validation distribution and has now informed development. The combined 1,400-scene score includes development scenes and is used for combination selection; it is not a new out-of-distribution test. We preserve these identities in every table. Checkpoints in the completed future screen use the fixed fourth-epoch endpoint and one fresh training seed. Paired comparisons share scenes and specified rollout seeds; scenario uncertainty does not include between-training-seed variation.

Outcomes and comparison scope. For $K = 14$ tasks with n_k episodes each, the primary score is

$$\mathrm{SR}(m) = \frac{1}{K}\sum_{k=1}^{K}\frac{1}{n_k}\sum_{i=1}^{n_k} y_{mki}, \qquad y_{mki} \in \{0,1\}. \tag{2}$$

Equal per-task counts make pooled and task-balanced SR identical here. We report counts as well as percentages. Paired differences use the same episodes and the uncertainty procedure attached to the source analysis. The completed future comparison corrects its twenty-seven overall contrasts as one family. Per-task findings are exploratory.

Matched internal studies share observation fields, training exposure, action interface, and execution settings. The external-method adaptation table instead compares complete

systems with different pretraining, training budgets, and decoders. It records a useful performance reference without identifying a particular architecture's effect. Contact-time and relative-pose errors, safety outcomes, and measured action age are reserved for the mechanism and deployment studies where their measurement interfaces are available. Missing readouts are not assigned zero. Representation-specific errors retain their native units; a latent loss is not ranked numerically against a metric pose error.

4. The SE(3)-WAM Design Study

The study represents a design by (H, Z, B, G): a history encoder H, future targets Z, backbone B, and information-flow program G. Figure 2 shows their roles. The completed screen changes Z under a fixed reference; subsequent stages examine H, B, and G and revisit retained alternatives. A single component's point-estimate ranking does not determine the final composition.

4.1. Q1: Information Access and History Encoding

The encoder maps timestamped history into conditioning tokens, $X_t = H(\mathcal{H}_t)$. Its design includes the physical window, temporal spacing, visibility, and motion coordinates. Current RGB, stacked RGB windows, codec motion vectors, optical flow, latent history, and per-frame query compression provide different descriptions of the same recent evolution. Encoding and information access are separate factors: supplying a ground-truth velocity changes what is known, while reorganizing the same visual samples changes how it is represented.

The current-pose study holds current target pose and eight robot-state samples fixed, removes target-history kinematics, and compares eight visual-history conditions plus a freshly trained GT-history reference. The object-motion track instead supplies seven intervals of target kinematics. This separation tests the conditional value of visual history rather than conflating temporal encoding with target localization. Window length and timestamp spacing are recorded independently of token count. A motion displacement also carries its time interval; a trajectory contains visibility rather than treating an absent point as stationary.

The diagnostic program complements task success with motion probes, matched-current history substitutions, occlusion, and sampling-gap tests. A probe measures what is decodable. A history substitution tests whether a policy responds to temporal information. Useful memory additionally requires better action outcomes under valid matched conditions. The necessary-current matching and closed-loop diagnostics remain part of the unfinished Q1 study.

4.2. Q2: Predicting Interaction-Relevant Variables

Q2 separates *scope* (target, context, or both), *variables* (pose, motion, visual features), and *granularity* (spatial and temporal allocation). The completed comparison includes no future target, scene-flow only, target-pose only, their combination, alternative scene/target descriptions, and additional future supervision. Figure 3 gives the central contrasts; Appendix D provides all fifteen conditions. Scene flow also covers target regions, so the scene-flow-only condition is not a context-only control. Pure role separation and allocation require additional matched comparisons.

For the rigid-target condition, let (p_t, R_t) be the current target pose in the fixed metric frame. At physical offset $\delta_h = 0.05h$ seconds, $h = 1, \ldots, 8$, the supervised variables are

$$d_h = p_{t+\delta_h} - p_t, \qquad \phi_h = \mathrm{Log}(R_{t+\delta_h} R_t^\top)^\vee. \quad (3)$$

Predicted poses reconstruct as $\hat{p}_{t+\delta_h} = p_t + \hat{d}_h$ and $\hat{R}_{t+\delta_h} = \mathrm{Exp}(\hat{\phi}_h^\wedge) R_t$. Translation and left-relative rotation are parameterized separately. The pose readout pools sixteen meta tokens and predicts eight cumulative transforms. Its valid-target loss combines squared translation error with 0.1 times squared rotation geodesic error. This geometric parameterization does not impose equivariance on the network.

Scene-flow supervision describes metric cumulative point displacement. Alternatives describe dense 2-D displacement, codec block motion, sparse scene trajectories with visibility, or future video latents. Target alternatives use pooled DINO features or target trajectories. The added-supervision conditions predict depth, DINO features, dynamic-region RGB, or their combination. Their labels have different units, support, and dimensionality. Replacing scene flow with dense 2-D motion consequently tests a representation package, not dimensionality alone. Likewise, pose plus sparse tracks is not by itself a matched proof that fine-target/coarse-context allocation is optimal.

The completed study uses these quantities as auxiliary training targets. Future labels and validity masks remain in their supervision branches; neither is an action input. Deterministic readouts and generated variables have distinct interfaces. To compare uniform and heterogeneous allocation directly, the remaining study must hold role information, temporal support, and an applicable token/compute budget common, then vary the allocation within a compatible representation. A target mask used to construct a training label does not establish deployable target assignment.

4.3. Reference Training and Action Generation

The common Q2 reference uses a four-layer transformer of width 768 with eight attention heads. A frozen DINOv3 ViT-L/16 encoder [24] provides 196 tokens from a 224×224 current image. The D backbone shares attention projections and uses separate feed-forward networks for world/meta, flow, and action streams. Action attends to meta and action tokens; it does not attend to predicted flow or flow hidden

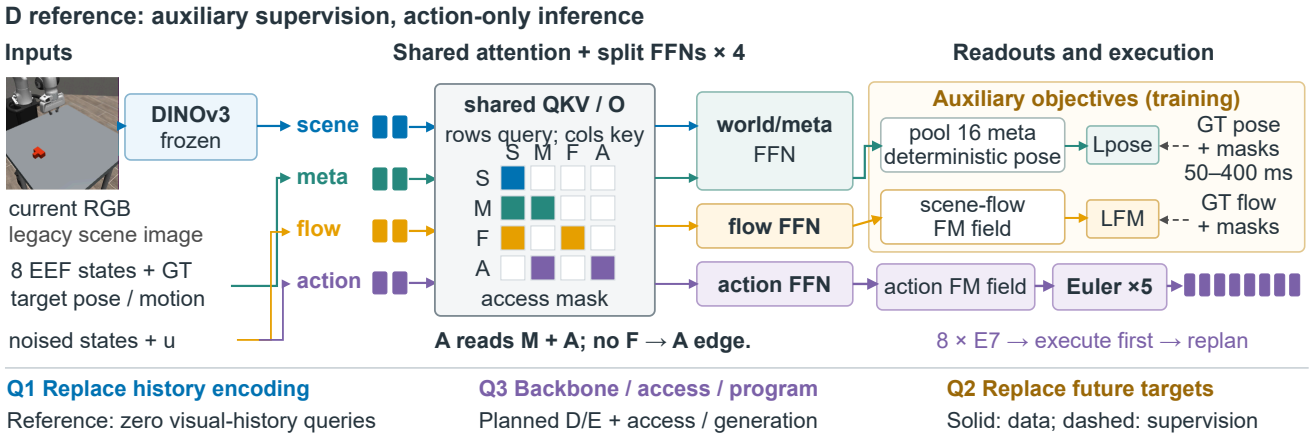


Figure 2. Implemented reference D, with P03's pose and scene-flow targets shown. Current RGB, robot history, and privileged target state form conditioning tokens. The mask gives query-to-key access for scene (S), meta (M), flow (F), and action (A). Within each of four layers, the streams share attention projections and use stream-specific FFNs. Pose is a deterministic readout; scene flow and action use flow-matching fields. Future labels and masks provide training supervision (dashed). Deployment solves only action with five Euler steps, executes the first E7 command, and replans. Depth and camera geometry are omitted for clarity; conditioning details are in Appendix B. Q2 changes the enabled targets; Q1/Q3 change history and computation.

states. Future objectives can still shape shared trainable features through their gradients.

Action generation uses flow matching [17]. With normalized action target a_1, source $a_0 \sim \mathcal{N}(0, I)$, and interpolation time $u \in [0, 1]$,

$$a_u = (1-u)a_0 + ua_1, \quad \mathcal{L}_A = \mathbb{E}\|v_\theta(a_u, u, X_t) - (a_1 - a_0)\|^2. \tag{4}$$

The complete objective adds only the enabled future losses, $\mathcal{L} = \mathcal{L}_A + \sum_{r \in Z} \lambda_r \mathcal{L}_r$. Masks restrict each loss to its valid support. A missing head contributes no measurement; raw total losses with different terms are not compared as one quality metric. Appendix B specifies the reference source distributions, coordinates, and implementation settings.

The action chunk contains eight seven-dimensional commands: translation increment, left-relative rotation vector, and a gripper command. At deployment, five Euler steps generate the action chunk, its first command is executed, and the policy replans. Conditioning caching is enabled; temporal and velocity caching are disabled. The training/evaluation translation bounds are respectively 0.05/0.10 m per axis. Only the action stream is solved online in the completed Q2 screen.

4.4. Q3: Backbone and Information Flow

The next backbone comparison holds selected H/Z and the reference action-only deployment program G_0 fixed. It compares D's stream-specific feed-forward organization with an adapted E world/action parameter organization. Both are freshly trained; the proposed active-backbone parameter difference is bounded by approximately 5%. This E is distinct from the object-input condition called E in earlier experiments and from the older native two-column E implementation. It is a planned matched backbone comparison, with no completed GPU result.

The subsequent seven-condition information-flow study holds $H/Z/B$ fixed: R0 removes the studied future loss; R1 retains it without cross-stream edges; R2 allows future-to-action access; R3 allows action-to-future access; R4 allows both; R5 generates future then action; R6 generates action then future. R1 also compares action-only and joint deployment of the same checkpoint. These contrasts distinguish supervision, information access, and generation programs. The sequential conditions use clean ground-truth first-stage values for second-stage training and predicted values at deployment, so their difference includes teacher forcing and its deployment mismatch.

The current generative interface is defined for scene flow. If representation selection retains sparse tracks or video latents, their generated states, normalization, noise, and loss must be specified before the information-flow comparison. Five joint evaluations and five-plus-five sequential evaluations match per-modality steps but not total compute. The planned comparison therefore reports actual calls, parameters, and same-device latency. Online substitutions of future, including oracle and valid swaps, are used only after a replaceable forecast interface exists.

5. Experiments

We report completed system references, future-supervision comparisons, combination selection, and the available history results. The backbone, information-flow, pretraining, and external studies are specified in Sec. 5.6. Tables distin-

Table 1. System references on the 1,260-scene complement (fourteen tasks, ninety scenes each). Training seed is 0. Methods retain their adapted input, initialization, and execution pipelines; budgets and fixed140 counts are in Appendix B. Scores compare complete systems, not matched architectures. The lower block receives GT target pose and interval kinematics.

System	Initialization / study condition	Successes /1,260	SR (%)
Adapted published systems			
ACT [41]	From scratch	424	33.65
Diffusion Policy [6]	From scratch	432	34.29
$\pi_{0.5}$ [23]	Official `pi05_base`	751	59.60
DynamicVLA [31]	Adapted native pipeline	363	28.81
OpenWAM-Alpha [28]	Official Alpha foundation	708	56.19
DynamicWAM [21]	Official full initialization	670	53.17
DreamVLA [39]: 20 epochs	Independent 20-epoch training schedule	1,000	79.37
40 epochs	Independent 40-epoch training schedule	1,003	79.60
Controlled-study systems with privileged target information			
P00: no future supervision	Current image; frozen DINO-L	791	62.78
P03: scene flow + pose	Current image; frozen DINO-L	786	62.38
I: scene tracks + pose	Full visual window; CNN; target RGB reconstruction	839	66.59

guish measured values from pending cells.

5.1. System References and Training Conditions

Table 1 places the study against adapted action and world–action models on the same fourteen-task evaluation scenes. The two DreamVLA schedules reach 1,000/1,260 and 1,003/1,260 successes. The study's combination candidate I reaches 839/1,260, and the no-future reference reaches 791/1,260. These are system-level observations. Candidate I and the reference have privileged target information; the adapted methods differ in initialization, training exposure, observation processing, and action execution. The table therefore motivates improving the current system while retaining the stronger reference, without attributing the gap to future representation.

Within the fifteen-condition future screen, all task networks start fresh with training seed 0 and run four epochs, 40,104 updates, and 641,664 sampled chunks at global batch 16. They use the fixed fourth-epoch checkpoint. Training is distributed over eight or sixteen devices with the same global batch; topology histories are retained in Appendix B. A single trained endpoint per condition supports paired scene comparisons but does not measure training-seed robustness. Current pose, motion history, action objective, and one-step receding-horizon execution are held common across this screen.

5.2. Q2: Which Future Supervision Helps?

Adding scene-flow and target-pose objectives. Figure 3 places the core contrasts in the full comparison. The no-future reference P00 succeeds in 791 scenes. Scene-flow P01, target-pose P02, and their combination P03 succeed in 782, 789, and 786 scenes, respectively. The combined objective does not show an overall improvement over either component or the reference in this setting. This result concerns auxiliary training with privileged motion input; online forecast consumption is a separate contrast.

Changing the predicted variables. Sparse scene tracks with target pose (P07) reach 807/1,260, and scene flow with target tracks (P09) reaches 808/1,260. Relative to P00, P07 gains 85 scenes and loses 69; P09 gains 91 and loses 74. Their raw two-sided exact McNemar p values are 0.227 and 0.213. The small net gains do not establish a reliable overall advantage. Non-rejection also does not establish equivalence: P12, for example, has the same total as P00 but exchanges 73 successes and 73 failures.

Dense 2-D future displacement with pose (P05) reaches 746/1,260, a decrease of 3.57 percentage points from P00 and 3.17 points from P03. The corresponding Bonferroni-adjusted p values over twenty-seven overall contrasts are 0.00844 and 0.01877. The comparison supports an adverse effect for this representation condition under the reported recipe. It does not isolate a universal 2-D versus 3-D ordering: P07 and P09 also use 2-D trajectories, with different support and target definitions.

Task variation and failure evidence. All fifteen conditions obtain zero successes on rigid-motion receptacle placement (`t2_se3`). On full constrained passage (`t3_full`), P00 and P06 obtain 3/90 and 4/90, with zero for the remaining conditions. Prediction changes have therefore not removed these shared task failures. P05's loss relative to P03 includes 27 fewer successes on rolling-sphere capture; its native `tray_contact` failures increase from 40 to 67 in that task. P09 gains on moving placement and replanning but

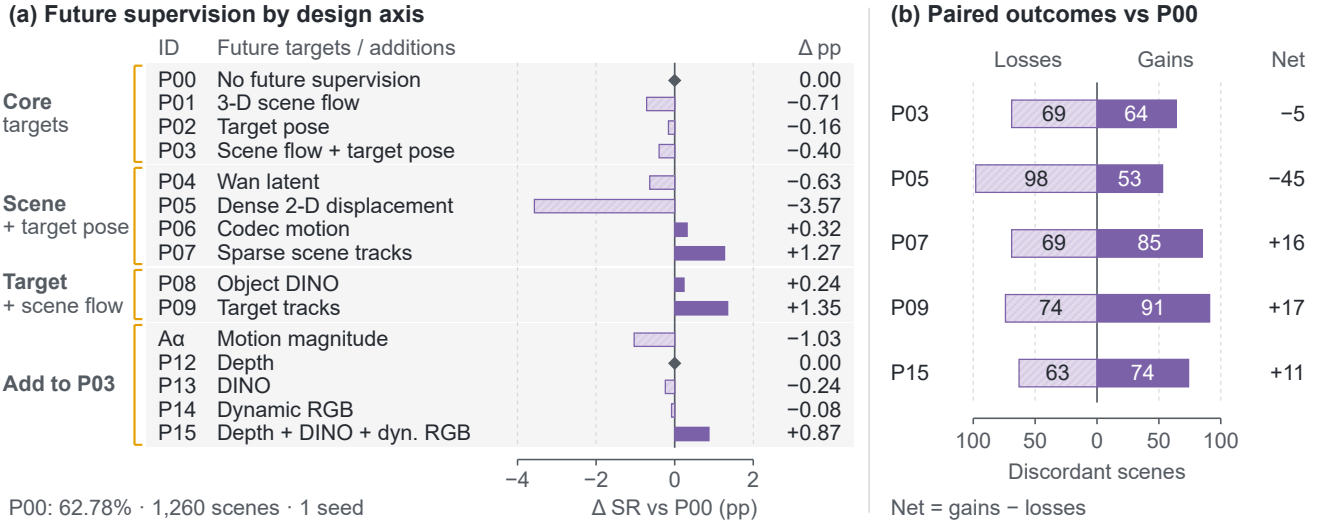


Figure 3. Future supervision at the fourth-epoch endpoint: one training seed, 1,260 paired scenes per condition (fourteen tasks, ninety each). (a) All fifteen conditions, grouped by the component changed; bars show success-rate differences from P00 (62.78%). (b) Selected contrasts expose scene-level changes: gain means P00 fails and the candidate succeeds; loss means the reverse. Net change is gain minus loss. P05 is adverse after the 27-contrast correction; P07/P09 do not establish an overall gain. No confidence intervals are shown. The separate fixed140 partition is in Appendix D.

loses on orientation-sensitive grasping. These observed failure codes and counts identify where to investigate; they do not identify a representation mechanism without an aligned intervention.

Development-set ordering. P01 has the highest fixed140 point estimate, while P09 has the highest 1,260-scene point estimate. The development-set ranking does not persist across the remaining validation scenes. Appendix D reports all conditions, paired gain/loss counts, and per-task results. The evidence favors retaining competitive alternatives for joint comparison instead of treating the first ranking as a final design principle.

5.3. Combining Future Targets with Visual Encoding

The twelve-condition combination study crosses P07/P09/P15 with DINO-S or a CNN encoder and with or without current-target RGB reconstruction. It uses a full visual window and privileged target-motion information. The predefined selection score is success over all 1,400 development/evaluation scenes. Candidate I, P07 with CNN and reconstruction, obtains 935/1,400 (66.79%), including 839/1,260 (66.59%) on the complement, and is selected for matched controls (Table 2).

Its 39-scene net gain over the same P07/CNN condition without reconstruction is concentrated: phase passage gains 44 scenes and orientation-sensitive grasping gains 17, while translational placement loses 10 and XYZ grasping loses 7. Both rigid-motion placement and full passage remain at zero for all twelve conditions. Thus the selection is useful for subsequent experimentation, but its overall score does not imply uniform task improvement. The matched P03 and P00 conditions using CNN and reconstruction are **[pending complete evaluation]**. Their outcomes are necessary to attribute any gain to the selected future target.

Table 2. Future–encoder combinations: successes /1,400 (the development selection score, including fixed140). Rec. denotes current-target RGB reconstruction. All conditions receive a full visual window and GT target motion. The P07/CNN/+rec. condition is selected for matched P00/P03 controls, which remain pending. Appendix D gives all candidate IDs and task counts.

	DINO-S		CNN	
Future supervision	No rec.	+rec.	No rec.	+rec.
P15: P03 + three targets	873	836	875	867
P07: scene tracks + pose	840	849	896	935
P09: flow + target tracks	820	830	829	870

5.4. Q1: Visual History Given Current Target Pose

Five of the nine current-pose study conditions have complete scores (Table 3). The current-image condition obtains 748/1,400 (53.43%); Window, Codec-MV, Flow, and PACT-Age obtain 662, 733, 720, and 634 successes. None of these completed history conditions exceeds the current-image control in point estimate. The remaining conditions and the fresh

Table 3. Visual history given current GT target pose and robot history. Historical target kinematics are removed. SR uses the 1,400-scene development total; Δ is relative to current image, in percentage points. Appendix C reports partition counts and four unfinished conditions separately.

Visual input	SR /1,400 (%)	Δ (pp)
Current image	53.43	0.00
Window	47.29	−6.14
Codec-MV	52.36	−1.07
Flow	51.43	−2.00
PACT-Age	45.29	−8.14

GT-history reference are pending, so the full Q1 comparison remains open.

This experiment preserves robot-state history and current GT target pose while removing target-history kinematics. It addresses visual motion information conditional on current localization. Its frozen DINO-L encoder, Alpha objective, and learning-rate schedule differ from the combination study. Consequently, the difference between their aggregate scores is not an estimate of a history-by-future interaction. The planned matched current-pose/GT-motion and current-image/history contrasts will test whether privileged motion reduces the incremental value of visual history. Completed probe extraction alone does not answer that question.

5.5. Execution and Learning Diagnostics

Future-screen public policy calls have same-device mean times of 177.071–184.212 ms and p95 times of 178.726–185.133 ms across conditions. Peak allocated memory is 1.571–1.617 GiB. These measurements include internal perception and action computation but exclude external oracle access, I/O, rendering, loading, and priming. They come from one fixed-order profiling round; small differences do not support a stable efficiency ranking. In particular, the simulation's 50 ms action interval is not evidence of 20 Hz wall-clock deployment.

The hard-task diagnostic also identifies a supervision/execution mismatch. In the historical full-passage set, all 800 demonstrations release at the final action, with release supervision at chunk position h_7 and none at the executed h_0. This is a coverage issue under one-step deployment, rather than absence of every release label. Short diagnostic training gives inconsistent successes, and the tested first-action weighting endpoints do not establish a solution. This evidence motivates checking supervision at the executed action position before explaining difficult-task failure through future representation alone. Appendix G gives the scope and negative results.

5.6. Remaining Studies and Final Model Selection

Backbone and information flow. After completing the combination and history controls, the matched D/E study uses three fresh training seeds per backbone, followed by the seven information-flow conditions in Sec. 4. The planned paired comparisons measure auxiliary supervision, online access, and generation order; their result slots are **[D/E and R0–R6 outcomes and cost]**. A retained representation is revisited after changing information flow, and the selected flow is applied to the previously unselected backbone. Necessary multi-seed and unused-scene confirmation precedes the final design. Appendix E gives the comparisons and missing generative interface.

Robot-data pretraining and independent adaptation. The confirmed design proceeds to training-recipe, model/data-scale, and compute studies. DROID preparation must validate geometry, timing, and selected targets; it does not directly provide object pose GT. A real-robot-data base independently initializes the simulation and real-robot post-training branches. Matched downstream budgets and equally budgeted recipe searches evaluate initialization. Results are **[pending]**; Appendix F gives the protocol.

External and deployment evidence. Single-arm DOMINO, static compatibility, and real conveyor manipulation test transfer after selection, using their native scores. Measured-delay tests report contact error, safety, action age, and end-to-end cost. The final SE(3)-WAM configuration and external results remain **[pending]** (Appendix F).

6. Discussion and Conclusion

Dynamic manipulation calls for a representation of the future that supports the interaction the robot must perform. The completed study shows that adding target and scene prediction objectives does not automatically improve a motion-informed policy: sparse trajectories give modest point-estimate gains, dense 2-D displacement is adverse under the tested recipe, and difficult tasks retain shared failures. Combination and history results further show why information access, training recipe, and execution must accompany a representation comparison.

The evidence currently covers fourteen simulated tasks, one training seed for the main screens, privileged target information, and synchronous execution. The fine-target/coarse-context hypothesis, useful online foresight, multi-seed design stability, and real-robot transfer require the specified comparisons. Final selection will use task-specific gains and losses, action-level interventions, and measured deployment cost.

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Supplementary Material

This supplement provides the complete experimental protocol, implementation, condition definitions, and measured results. Planned comparisons are identified where their outcomes remain unavailable. The same appendix sources are included in the standalone supplement and the complete English reading edition.

Appendix	Contents	Main
A	Tasks, data, information, scoring	Sec. 3
B	Implementation and system budgets	Secs. 4–5
C	History conditions and results	Sec. 5.4
D	Future targets, statistics, combinations	Secs. 5.2–5.3
E	Backbone and information flow	Secs. 4.4, 5.6
F	Pretraining, deployment, external tests	Sec. 5.6
G	Failures and qualitative evidence	Sec. 5.5

A. Tasks, Data, and Evaluation

A.1. The Reported Fourteen-Task Suite

The completed studies use the migrated DB0 v0.6 demonstration pool and its fourteen-task evaluation suite. Table S1 lists the task identifiers used in the result arrays. These identifiers are retained so that results cannot be silently pooled into a changed task taxonomy. The task-balanced score includes every task, including zero-success conditions.

Table S1. Task identifiers and the interaction they exercise. These are the reported evaluation tasks, not a count of new environments introduced by this manuscript.

Identifier	Interaction
grasp	Reference object grasping
t1_belt	Moving conveyor target
t1_occ	Moving target with physical occlusion
t1_so3	Orientation-sensitive target grasping
t1_xyz	Translating target grasping
t2_se3	Placement into a rigidly moving receptacle
t2_trans	Placement into a translating receptacle
t3_full	Full constrained-passage task
t3_phase	Phase-dependent passage
t4_can	Roll/slip can capture
t4_slider	Sliding-object capture
t4_sphere	Rolling-sphere capture
t4_sphere_tabletop	Tabletop sphere capture
t5_replan	Post-impact replanning

The common training pool has 11,200 episodes, 800 per task. The internal representation screen uses 350 validation episodes; adapted systems retain their own validation settings (including 1,400 for DreamVLA/OpenWAM). The future screen samples 160,416 chunks per epoch. Stored episode count and effective chunk exposure measure different aspects of training. Later fresh-data production retains its own data identity and does not retroactively change the source of these results. Motion-only P0 prediction has no corresponding action-learning score in the fourteen-task table.

Success is read from the native task evaluator after its ordered physical requirements. A final pose or incidental contact is not substituted for the evaluator outcome. Native failure categories are boundary, safety, task violation, and timeout. These categorical outcomes are useful for localization of failures; the word safety is an evaluator category rather than a certified real-robot safety guarantee. The exact numerical event thresholds and evaluator configuration for an independently distributable release remain **[release configuration to attach]**. Historical grasp thresholds from earlier manuscripts are not substituted for those actual settings.

A.2. Partitions and Statistical Interpretation

The 140-scene development set has ten scenes per task. Its disjoint complement has ninety per task, totaling 1,260. The future screen reports the two partitions separately. The system-reference study also keeps them separate. The combination screen explicitly selects on all 1,400 scenes while retaining both component counts. Thus its selection score includes development data, and the 1,260 complement is now part of the project's development evidence. Neither membership disjointness nor a different rollout seed makes it a final blind test.

The future study uses each condition's own fourth-epoch endpoint and one training seed. Across arms, the scene, source seed, and inference seed are paired. There are 18,900 accepted outcomes, 1,260 per arm; missing scenes and infrastructure-error exclusions are zero. Inference seeds span 20261058–20262317 for the designated selection indices. Matching an inference seed specifies rollout randomness but does not imply that unlike training graphs consumed identical random-number sequences.

For two policies on the same scenes, let b count candidate successes/reference failures and c candidate failures/reference successes. Then

$$\Delta\mathrm{SR} = 100(b - c)/1260 \qquad \text{(S1)}$$

in percentage points. The report uses the exact two-sided McNemar test on discordant pairs. Fourteen comparisons to P00 and thirteen to P03 form the twenty-seven-comparison family; adjusted values use $\min(1, 27p)$. The 378 task-level comparisons remain exploratory. No confidence interval is invented for a source that reports none, and scenario-level tests do not estimate training-seed variation.

A.3. Inputs and Time

The completed future screen uses one current fixed-camera RGB view, exact depth/camera geometry, robot-state history, and privileged object information. A simulator field stored beside an image is not automatically a deployable sensor observation. Future target states and label-validity masks supervise their respective branches and are excluded from the action condition. A current-object-pose oracle and a future oracle are distinct information conditions.

Physical high-level steps are separated by 50 ms. Forecast labels cover 50–400 ms relative to the source observation. Synchronous evaluation pauses the environment for policy computation, so it cannot measure the consequence of wall-clock observation age. Later fixed-delay and real-delay evaluations must retain their own time traces and report whether physics continues while the policy computes. Contact-dependent future labels require an audit of the generating action and valid prediction support; completion of that audit is **[pending for the final online-generation protocol]**.

B. Reference Implementation and Budgets

B.1. Actual Representation-Screen Configuration

The reference D model uses `se3wam_v2/split_ffn`, four transformer layers, width 768, eight attention heads, and factorized attention. DINOv3 ViT-L/16 is frozen [24]. A 224×224 agent-view image produces 196 current visual tokens. H0V has zero visual-history query tokens. The information track also supplies exact depth and camera geometry. Robot input retains eight EEF7 states at 20 Hz. Eight object tokens encode current GT pose and seven motion intervals using 25 numeric statistics and pose/interval/acceleration validity flags.

The object statistics include position, rotation vector, interval translation and relative rotation, elapsed time, linear/angular velocity, and linear/angular acceleration. Their source is simulator truth in this track. The current-pose history track removes historical object poses and all derived interval kinematics while preserving the current pose and robot history. This is a different information condition, not a different name for the same input.

D shares attention QKV/output projections and splits feed-forward networks by world/meta, flow, and action stream. Scene tokens access scene tokens; meta tokens access scene and meta; flow accesses scene and flow; action accesses meta and action. Consequently, flow supervision can change shared weights without an online flow-to-action path. The deterministic pose head pools sixteen final meta tokens. Calling the model world–action does not change these actual access relations.

B.2. Coordinates, Losses, and Generation

Target future translation is cumulative displacement in the fixed metric frame. Rotation is the left-relative vector $\phi_h = \mathrm{Log}(R_{t+\delta_h} R_t^\top)^\vee$. For valid targets $\mathcal{V}$, the pose loss is

$$\mathcal{L}_{\mathrm{pose}} = \frac{1}{|\mathcal{V}|} \sum_{h \in \mathcal{V}} \big[\|\hat{d}_h - d_h\|_2^2 + 0.1\, d_{\mathrm{SO}(3)}(\hat{R}_{t+\delta_h}, R_{t+\delta_h})^2 \big]. \tag{S2}$$

The external weight multiplying this loss is separate from the internal rotation coefficient. A full twist exponential would couple translation to rotation through its Jacobian and is not the implemented parameterization. The reported configuration does not include the former draft's heteroscedastic pose likelihood, uncertainty bridge, symmetry-minimized loss, or pose–flow consistency objective.

The action has eight E7 steps: base-frame translation increment, left-relative rotation vector, and gripper command. Translation and rotation training bounds are 0.05 m and 0.5 rad per axis. The inherited evaluation translation limit is 0.10 m per axis. Action flow matching uses a unit Gaussian in training-normalized action coordinates. Scene flow uses a metric Gaussian source with standard deviation 0.02 m. Deterministic auxiliary readouts, including pose, do not consume flow-matching source noise and time.

Only the action generator runs during Q2 deployment, using five Euler steps. Conditioning caching is on; temporal/velocity caching is off. Each replan executes one command. Trainable initialization is fresh for seed 0 rather than inherited from a trained common-reference checkpoint. Global batch is 16. Four epochs produce 40,104 updates and 641,664 chunks. P00, P02, and P05 finish with an eight-device topology; other arms finish with sixteen. Some retained runs have recorded topology transitions. These histories preserve the declared exposure and endpoint but do not establish bitwise equivalence across topology.

B.3. Alternative Target Definitions

P04 uses a frozen Wan video VAE target. Nine RGB input frames produce three temporal latents in the target construction; this is not eight independent future latent frames. P05 uses dense cumulative 2-D displacement. P06 uses real codec block motion vectors with reference, direction, and validity. P07 uses sparse scene tracks and visibility, P08 pooled object DINO features, and P09 target tracks and visibility. P08/P09 adapt mechanisms rather than reproduce an entire published model.

Aalpha adds motion-magnitude MSE with weight 0.5 while preserving P03's Alpha-head initialization. P12, P13, and P14 add future depth, spatial DINO, and dynamic-region RGB supervision. P15 combines these three at their individual weights. A non-existent head has an inapplicable readout, not a zero error. The combination cannot identify

every pairwise interaction. Raw total losses across these objectives are not commensurate.

B.4. System-Reference Training

ACT and Diffusion Policy train from scratch for 50,000 updates at global batch 32. The $\pi_{0.5}$ adaptation starts from `pi05_base` and trains 30,000 updates at global batch 256 [23]. DynamicVLA uses 50,000 updates at global batch 44. OpenWAM uses a fresh fine-tuning run from the official Alpha foundation, for five epochs, 34,490 updates at global batch 192, with its declared foundation components. DynamicWAM uses its official full initialization and 40,000 updates at global batch 128 [21, 28]. DreamVLA's 20-epoch and 40-epoch schedules use independent training runs, 48,220 and 96,440 updates at global batch 512; the shorter endpoint is not the midpoint checkpoint of the longer schedule [39].

Executors also differ: ACT uses temporal aggregation, Diffusion Policy executes eight actions, $\pi_{0.5}$ ten, and DynamicWAM sixteen before replanning. The internal representation screen executes one. System-level interpretation retains these differences. Matched-budget comparisons must also report each system's inputs, initialization, and execution configuration.

C. History Encoding and Conditional Information

C.1. Completed and Outstanding Conditions

The current-pose study compares eight history encodings plus a fresh GT-history reference. Every visual-history condition preserves current GT target pose and eight robot-state samples, while historical object pose and derived kinematics are masked. Its reference future supervision is held fixed. Completed conditions are current image, Window, Codec-MV, Flow, and PACT-Age. Flow12D, Wan, PACT-Uniform, and the fresh GT-history reference have not yet produced a complete accepted comparison table. PACT-Age distributes fifty-six history queries from older to newer frames as $[2, 4, 6, 8, 10, 12, 14]$; PACT-Uniform uses seven groups of eight. These adapt PACT-WAM's temporal allocation principle [19] through per-frame query compression, rather than reproducing its complete model.

Each of the five completed conditions has probe extraction over 350 episodes and 1,050 windows with 306 valid readouts. This establishes availability of the diagnostic arrays. The manuscript has not turned those arrays into a reported probe-effect estimate. Their final matched metrics and complete nine-condition input/noise/label audit are **[pending]**. The observed point estimates support continuing the comparison without presuming a visual-history gain.

C.2. Recipe Differences and Required Contrasts

This study uses frozen DINO-L, retains the original Alpha objective, and has a cosine schedule over the first 10,026 updates followed by three epochs at 2×10^{-5}. The combination study uses DINO-S or CNN, disables Alpha, and applies cosine scheduling across all 40,104 updates. Its stronger score therefore cannot be subtracted from the history-study score to estimate a visual-history effect.

The planned core comparison crosses current-pose versus GT-motion input with current-image versus visual-history encoding under a common recipe. Multiple training seeds and paired scenes are needed to estimate the conditional history benefit. A motion probe and an action intervention answer different questions: recoverability of velocity does not demonstrate that the policy used velocity to improve interaction.

C.3. Matched-Current and Occlusion Diagnostics

History substitutions require matching the current target pose, robot state, view, and task condition while changing valid preceding motion. The matching tolerance, group counts, and history-necessary subset acceptance remain **[pending]**. A substitute that changes the current state tests an additional intervention. Occlusion tests must report visibility and duration in physical time, while sampling-gap tests preserve the meaning of displacement by retaining elapsed time. The planned outputs link motion decoding, action change, contact outcome, and unconditional task success. No-contact failures remain in the success denominator rather than receiving zero contact error.

D. Future Supervision and Combination Selection

D.1. All Fifteen Future Conditions

All conditions in Table S4 have action supervision. Sceneflow coverage includes target regions, so P01 is not a pure context-only condition. P03 versus P01/P02 compares target-pose and scene-flow objectives under the common action path; it does not isolate an exclusive target/context spatial partition. Main-paper Fig. 3 visualizes all fifteen point estimates without adding confidence intervals absent from the source analysis.

P00/P01/P02/P03 test no future objective, scene flow, pose, and their combination. P04–P07 replace the scene target while retaining pose. P08/P09 replace the object target while retaining scene flow. Aalpha and P12–P15 add supervision to P03. Full target definitions and source units are given in Appendix B. None of these conditions changes deployment into online forecast consumption.

Table S2. Training exposure and development counts for all system-reference endpoints. Batch is global; all endpoints use training seed 0. Initialization and execution are part of each adapted system, not matched factors. An unspecified field requires a configuration audit before a matched-budget comparison. The two DreamVLA endpoints are independent training schedules. The 1,260-scene counts appear in the main system table.

System	Updates	Batch	Success /140	Execution / initialization note
ACT	50,000	32	50	Scratch; temporal aggregation
Diffusion Policy	50,000	32	54	Scratch; execute 8 actions
$\pi_{0.5}$	30,000	256	89	`pi05_base`; execute 10
DynamicVLA	50,000	44	24	Native continuous pipeline; initialization audit pending
OpenWAM-Alpha	34,490	192	76	Official Alpha foundation; five-epoch fine-tuning
DynamicWAM	40,000	128	72	Official full initialization; execute 16
DreamVLA: 20 epochs	48,220	512	112	Independent shorter schedule
40 epochs	96,440	512	114	Independent longer schedule
Study P00	40,104	16	88	Fresh task network; frozen DINO-L; execute 1
Study P03	40,104	16	93	Same reference recipe as P00
Study I	40,104	16	96	Fresh CNN condition; full visual history; execute 1

Table S3. Available history results. The total is a development summary, not a separate held-out test. Dashes denote unfinished comparisons.

Condition	/140	/1,260	/1,400	SR (%)
Current image	80	668	748	53.43
Window	69	593	662	47.29
Codec-MV	77	656	733	52.36
Flow	76	644	720	51.43
PACT-Age	65	569	634	45.29
Pending visual-history conditions (same information)				
Flow12D	—	—	—	—
Wan	—	—	—	—
PACT-Uniform	—	—	—	—
Pending reference with privileged target history				
Fresh GT history	—	—	—	—

D.2. Paired Outcomes and Task Heterogeneity

Table S5 preserves gains, losses, and raw exact McNemar probabilities against both reference conditions. P05's adjusted probabilities are 0.00844 versus P00 and 0.01877 versus P03. The remaining positive overall differences are not established advantages. The absence of a detected difference is not an equivalence test.

Table S6 gives every task count with equal denominators. The strongest task-specific changes are exploratory. For P05 versus P03, rolling-sphere success falls from 50 to 23 while `tray_contact` failures rise from 40 to 67. Conversely, post-impact replanning rises from 54 to 60. P09's gains in translational placement and replanning coexist with fewer orientation-sensitive grasps. These observations narrow the conditions to investigate, without treating a native failure code as a learned-mechanism explanation.

The lowest count of native safety failures is P06's 250, but its 96 timeouts are the highest. P15 has only 37 timeouts and 343 safety failures. A favorable change in one category must be interpreted alongside all outcomes. Every policy failure remains in the 1,260-scene denominator.

D.3. Combination Selection

The next screen crosses three future choices, two visual encoders, and current-target RGB reconstruction on/off. It preserves a full visual window and privileged target-motion input, disables Alpha and late injection, and trains each condition fresh for four epochs with the full-run cosine schedule. Its twelve conditions are complete; matched P03/P00 controls of the selected configuration remain unfinished.

Candidate I obtains 935/1,400 by the predefined selection score. Against J, the same P07/CNN condition without reconstruction, the paired counts are 817 joint successes, 118 I-only successes, 79 J-only successes, and 386 joint failures. Its net 39-scene gain is distributed unevenly over tasks, as Table S8 shows. The selected candidate is carried forward for testing; the selection does not establish stable performance across training seeds.

D.4. Granularity and Prediction-to-Action Evidence

A future comparison must separate role coverage, variable choice, spatial allocation, and compute. The fifteen-target screen changes target packages, not only allocation at equal budget. A direct fine-target/coarse-context test still requires a compatible uniform reference with the same identity information, horizon support, and declared token/compute budget. Its result is **[pending controlled granularity comparison]**.

Physical readouts should compare target translation and

Table S4. All fifteen future-supervision conditions, grouped by the objective changed. Each uses one training seed, four epochs, common privileged object-motion input, and action-only deployment. The 140 development scenes and 1,260-scene complement remain separate. Scene flow includes target regions.

		Development	Disjoint complement	
ID	Future supervision	Success / 140	Success / 1,260	SR (%)
Reference and component objectives				
P00	No future supervision	88	791	62.78
P01	3-D scene flow	95	782	62.06
P02	Target pose	92	789	62.62
P03	Scene flow + target pose	93	786	62.38
Scene replacements: target-pose objective retained				
P04	Wan latent + pose	92	783	62.14
P05	Dense 2-D displacement + pose	87	746	59.21
P06	Codec motion + pose	86	795	63.10
P07	Sparse scene tracks + pose	90	807	64.05
Target replacements: scene-flow objective retained				
P08	Scene flow + object DINO	89	794	63.02
P09	Scene flow + target tracks	88	808	64.13
Additional objectives: scene flow + target pose retained				
A_α	P03 + motion magnitude	89	778	61.75
P12	P03 + depth	90	791	62.78
P13	P03 + DINO	92	788	62.54
P14	P03 + dynamic RGB	89	790	62.70
P15	P03 + depth / DINO / dynamic RGB	94	802	63.65

rotation at each horizon; flow and tracks require valid correspondences and visibility. Their association with contact error is a separate analysis. The accepted source contains readout arrays, but this manuscript does not substitute flow-matching validation loss for freely generated physical error. The aligned prediction/contact figure requires **[generated predictions, actual states, and contact events]**. If a future oracle is introduced, it must use the same action interface and target times. Any oracle-gap ratio is reported only with a supported, non-negligible denominator.

E. Backbone, Information Flow, and Cross-Checks

E.1. D/E Backbone Comparison

BACKBONE0 is prepared but has no completed GPU training, policy evaluation, or measured-cost result. Its input is the confirmed complete history/future configuration, retaining the reference G_0 program where action does not read predicted future. D uses split feed-forward networks and shared attention projections. The adapted E organizes world and action parameters separately within the current model interface. The old native E and the object-input condition E retain separate identities.

The planned two-arm comparison trains fresh seeds 0, 1, and 2, with the same initialization sources, data exposure, optimizer schedule, and endpoint. If the current four-epoch recipe is retained, there are 240,624 updates across six runs and 8,400 evaluation episodes at 1,400 per endpoint. These are planned counts. The active-backbone parameter difference is targeted within 5%; frozen frontend parameters and their runtime costs are recorded separately. Matching exposure or parameter count does not match training FLOPs.

The predeclared practical SR and non-inferiority margins are two percentage points, with a 10% p95 latency benefit considered for cost selection. Intervals must support the corresponding superiority, equivalence, or non-inferiority criterion before that label is used. If three seeds are inconclusive, both arms may be expanded to five under the existing plan. Same-device public-policy timing includes internal perception/action computation and excludes external rendering and oracle access. Selection and uncertainty are **[pending]**.

E.2. Seven Information-Flow Conditions

R0 versus R1 tests auxiliary supervision under action-only deployment. R1's two deployment modes test the cost and effect of generating an unused future from the same checkpoint. R1–R4 examine information access. R2 versus R5 and R3 versus R6 compare synchronous and sequential programs. Sequential second stages train on clean first-stage

Table S5. Paired outcomes over the same 1,260 scenes. Gain means candidate success/reference failure; loss means candidate failure/reference success. Probabilities are exact two-sided McNemar tests. Adjustment uses the full family of fourteen P00 and thirteen P03 contrasts: $p_{\mathrm{adj}} = \min(1, 27p)$. P03 against itself is inapplicable.

	P00: no future supervision				P03: scene flow + target pose			
ID	Gain	Loss	Raw p	Adjusted p	Gain	Loss	Raw p	Adjusted p
Reference and component objectives								
P01	80	89	0.538427	1.000000	72	76	0.805321	1.000000
P02	61	63	0.928492	1.000000	61	58	0.854630	1.000000
P03	64	69	0.728854	1.000000	—	—	—	—
Scene replacements: target-pose objective retained								
P04	74	82	0.575315	1.000000	71	74	0.868159	1.000000
P05	53	98	0.000312	0.008435	47	87	0.000695	0.018771
P06	81	77	0.811457	1.000000	68	59	0.477922	1.000000
P07	85	69	0.226651	1.000000	81	60	0.091769	1.000000
Target replacements: scene-flow objective retained								
P08	75	72	0.869050	1.000000	59	51	0.504685	1.000000
P09	91	74	0.212786	1.000000	79	57	0.071353	1.000000
Additional objectives: scene flow + target pose retained								
A_α	60	73	0.298071	1.000000	53	61	0.512255	1.000000
P12	73	73	1.000000	1.000000	63	58	0.716294	1.000000
P13	70	73	0.867250	1.000000	60	58	0.926704	1.000000
P14	71	72	1.000000	1.000000	62	58	0.784328	1.000000
P15	74	63	0.392988	1.000000	63	47	0.152338	1.000000

labels and deploy on first-stage predictions. The corresponding train/test mismatch belongs in the interpretation.

The current generative state is scene flow with shape $[B, S, 196, 8, 3]$; pose, tracks, depth, DINO, and RGB can be deterministic auxiliary readouts. A selected sparse-track or Wan representation requires its own generated-state, normalization, noise, and loss definition before this experiment can run. Replacing it with scene flow would change the question. This interface is **[pending final representation binding]**.

The planned three seeds yield twenty-one trained configurations and twenty-four deployment configurations, including R1's second mode; at 1,400 scenes each, that is 33,600 rollout episodes. Alpha is fixed to zero. Per-modality Euler5 is common, but joint inference uses five combined calls and sequential inference uses five plus five. Actual calls, latency, parameters, training exposure, and valid readouts are required alongside SR. The planned counts do not imply completed runs.

E.3. Mechanisms and Final Design

An independently replaceable future interface permits oracle, valid matched-current swap, time-shift, or calibrated-noise diagnostics. Each intervention holds perception, robot state, and action-source noise fixed. Changes in action alone show sensitivity; evidence for useful forecast use also requires task and safety outcomes. Removing a branch at inference and retraining without its loss answer different questions.

After selecting information flow, the study revisits retained future representations and applies the selected flow to the unselected backbone. History choices are revisited when the future target changes. Only the necessary comparisons are repeated; previously valid matched endpoints can be reused. Ranking reversals change the conditional design choice. Multi-seed and scenes not used for selection are required before passing a complete (H, Z, B, G) design to pretraining. The selected configuration, cross-check estimates, and mechanism readouts remain **[pending]**.

F. Pretraining, Cost, and External Evaluation

F.1. Measured Policy Cost and Physical Delay

The future-screen policy call means span 177.071–184.212 ms; p95 spans 178.726–185.133 ms; peak allocated memory spans 1.571–1.617 GiB. These are same-device, fixed-order measurements including internal perception and action computation. Loading, external oracle reads, I/O, rendering, and priming are excluded. One measurement round supports this range, not a robust ranking of differences of a few milliseconds.

For deployed commands, action age is the difference between sensor capture and command application. The planned

Table S6. Future-study successes: ninety scenes per task and 1,260 per condition. Columns follow the task families in Table S1; all tasks are retained, including zero-success tasks. Occl. denotes physical occlusion; Trans. denotes translating-receptacle placement; Tabletop denotes tabletop sphere capture.

	Reference	T1: target motion				T2: placement		T3: passage		T4: capture				T5	
ID	Grasp	Belt	Occl.	SO(3)	XYZ	SE(3)	Trans.	Full	Phase	Can	Slider	Sphere	Tabletop	Replan	Total
Reference and component objectives															
P00	85	90	90	75	76	0	8	3	15	90	69	45	86	59	791
P01	84	90	90	64	69	0	18	0	11	90	74	46	90	56	782
P02	87	90	90	78	73	0	7	0	0	90	71	59	89	55	789
P03	86	90	88	75	77	0	7	0	6	86	80	50	87	54	786
Scene replacements: target-pose objective retained															
P04	83	85	84	73	76	0	20	0	0	90	72	54	89	57	783
P05	84	89	90	66	75	0	10	0	0	90	71	23	88	60	746
P06	86	88	89	79	70	0	6	4	2	89	72	60	85	65	795
P07	82	90	90	78	71	0	17	0	0	89	79	55	90	66	807
Target replacements: scene-flow objective retained															
P08	84	90	90	71	81	0	13	0	0	90	77	60	90	48	794
P09	84	90	89	65	78	0	21	0	0	90	77	59	90	65	808
Additional objectives: scene flow + target pose retained															
A_α	86	90	90	72	75	0	3	0	0	85	73	55	90	59	778
P12	83	90	90	76	76	0	5	0	0	89	77	57	90	58	791
P13	86	90	90	80	73	0	6	0	0	88	78	54	90	53	788
P14	85	89	90	72	79	0	5	0	0	90	80	53	88	59	790
P15	87	90	90	79	81	0	11	0	0	90	75	54	90	55	802

Table S7. Native evaluator failure counts out of 1,260 scenes per condition, in the same design groups as the success tables. Safety is the evaluator category, not a certified safety measure. Every failure remains in the success-rate denominator.

ID	Future supervision	Boundary	Safety	Task viol.	Timeout	Total
Reference and component objectives						
P00	No future supervision	84	297	17	71	469
P01	3-D scene flow	62	324	23	69	478
P02	Target pose	77	331	17	46	471
P03	Scene flow + target pose	76	295	51	52	474
Scene replacements: target-pose objective retained						
P04	Wan latent + pose	79	310	41	47	477
P05	Dense 2-D displacement + pose	87	356	14	57	514
P06	Codec motion + pose	96	250	23	96	465
P07	Sparse scene tracks + pose	47	329	20	57	453
Target replacements: scene-flow objective retained						
P08	Scene flow + object DINO	61	322	42	41	466
P09	Scene flow + target tracks	67	299	35	51	452
Additional objectives: scene flow + target pose retained						
A_α	P03 + motion magnitude	57	315	65	45	482
P12	P03 + depth	62	327	40	40	469
P13	P03 + DINO	71	326	27	48	472
P14	P03 + dynamic RGB	70	322	18	60	470
P15	P03 + depth / DINO / dynamic RGB	59	343	19	37	458

Table S8. Combination-study successes, grouped by future target and visual encoder. DINO is frozen DINO-S; CNN is trained from scratch. Minus/plus denotes current-target RGB reconstruction off/on; parentheses retain source condition IDs. Each task has one hundred scenes, including development scenes, and each condition has one training seed. Totals use the predefined 1,400-scene selection pool.

Task	P15: additional targets				P07: scene tracks				P09: target tracks			
	DINO− (B)	DINO+ (A)	CNN− (D)	CNN+ (C)	DINO− (H)	DINO+ (G)	CNN− (J)	CNN+ (I)	DINO− (L)	DINO+ (K)	CNN− (N)	CNN+ (M)
Reference grasp												
Grasp	90	89	91	92	90	92	87	90	79	85	77	87
T1: target motion												
Belt	100	100	99	100	100	100	100	97	99	100	90	99
Occlusion	100	100	99	99	99	100	100	97	96	99	82	98
SO(3)	86	71	80	88	75	87	69	86	69	62	64	78
XYZ	58	58	82	75	73	75	89	82	44	56	82	84
T2: receptacle motion												
SE(3)	0	0	0	0	0	0	0	0	0	0	0	0
Translation	8	4	6	5	9	1	14	4	4	5	9	14
T3: constrained passage												
Full	0	0	0	0	0	0	0	0	0	0	0	0
Phase	0	0	0	0	0	0	28	72	0	0	0	2
T4: object capture												
Can	99	99	99	98	100	99	99	100	97	99	96	99
Slider	88	85	88	91	74	83	86	81	84	84	88	88
Sphere	71	67	67	63	47	40	53	56	68	68	62	64
Tabletop sphere	100	100	100	99	99	100	95	98	100	100	100	100
T5: post-impact replanning												
Replan	73	63	64	57	74	72	76	72	80	72	79	57
Total / 1,400	873	836	875	867	840	849	896	935	820	830	829	870

Table S9. Planned Q3 conditions; no measured result is assigned to these rows. F denotes a generated future state and A an action state. Arrows specify information access, independently of whether the future loss is enabled. R1 has two deployments of the same trained checkpoint. Sequential second stages use clean ground truth in training and generated first-stage values at test time.

ID	Future loss	$F \to A$	$A \to F$	Cross-stream training input	Inference order
R0	No	—	—	None	A only
R1	Yes	—	—	None	A only
				Same checkpoint	Joint F and A
R2	Yes	Yes	—	Synchronous noisy states	Joint F and A
R3	Yes	—	Yes	Synchronous noisy states	Joint F and A
R4	Yes	Yes	Yes	Synchronous noisy states	Joint F and A
R5	Yes	Yes	—	Clean F for second-stage A	F, then A
R6	Yes	—	Yes	Clean A for second-stage F	A, then F

deployment profile separates capture, availability, preprocessing, inference, queuing, issue, and application. It records hardware, precision, batch size, warm-up, synchronization, and timed sample counts. Fixed-delay simulation tests a controlled lag; wall-clock execution tests the actual combined system. The final same-device profiles, deadline misses, and action-age distributions are **[pending]**. A policy's public function latency must not be labeled full environment end-to-end latency when parts of that path are excluded.

F.2. Recipe and Robot-Data Base

After the design cross-checks, the study fixes input modalities, history support, future targets/horizons, information flow, and action semantics. The next stage optimizes the existing loss weights and training recipe, and changes depth/width within the chosen architecture for scale compar-

Table S10. Evidence slots for the final model. All are planned and currently unmeasured in this manuscript.

Track	Evidence question and required output
Unused simulation scenes	Does selection generalize? Multi-seed SR and uncertainty on unused scenes.
Robot-data base	Does initialization help? Paired adaptation, data/compute scale, and cost.
DOMINO single arm	Does the design transfer to other dynamic tasks? Native score and adaptation details.
Static compatibility	Are static skills retained? Native score with common tuning.
Real conveyor	Does the integrated system work physically? Trial counts, success, and interventions.
Measured delay	Does performance survive actual action age? Timing, safety, and success.

ison. Changing the representation or adding an information path belongs to a new design comparison rather than an optimizer ablation.

DROID supplies real robot demonstrations [13]. Preparation checks sensor timing, calibration, depth, command versus measured motion, gripper semantics, and training-only normalization. It does not provide object pose GT; the target-pose pipeline must be separately validated for any selected pose-based objective. Only targets used by the selected design and necessary ablations need to be produced. Valid label masks and rejected-action rates are part of the data report, not silently filled values. This stage is **[planned]**.

The formal base is trained on real robot data. Two independent post-training branches start from the same base: one for simulation and one for real-robot tasks. Simulation-from-scratch experiments supply recipe experience and a strong reference, while their weights do not initialize the formal robot-data base or the real-robot branch. Comparisons with and without pretraining share downstream data and exposure, and additionally allow equally budgeted recipe searches. They report the pretraining cost explicitly. Data, model, and compute scale have separate axes; unrun settings are not points on a scaling curve.

F.3. Planned External and Real-Robot Evidence

Single-arm DOMINO [8] tests dynamics under a different task/data protocol. Static compatibility uses an explicitly selected native suite such as LIBERO [18]. Real conveyor manipulation evaluates perception, prediction, action, and timing together. The final phase reuses completed pretraining, scaling, and independent adaptation results, then fills uncovered object, scene, motion, and delay conditions. It does not repeat an already completed scaling matrix.

The real-robot protocol must specify objects, speed/delay strata, trial allocation, ordering, calibration, safety termination, and human intervention. The old budget of 180 trials is not a completed count or an adopted current sample size. External native scores are not pooled into the fourteen-task simulation SR. Final training scale, robot trials, and real-world performance are **[pending]**.

G. Failure Evidence and Qualitative Experiments

G.1. Supervision at the Executed Action Position

The hard-task investigation finds that all 800 historical full-passage demonstrations release only at the final action. Release is represented at chunk position h_7 but has no h_0 supervision in that set, whereas the deployment program executes h_0. Thus the problem is action-position coverage. It cannot be summarized as completely missing release labels.

Short tail-window training produces 2/16 successes in one diagnostic validation and 1/6 in a follow-up selected-scene check; this does not establish reliable full-task learning. For phase passage, three seeds after 256 updates give 0/96, 12/96, and 26/96; at 1,024 updates all three give 0/96. Six first-action-weighting endpoints total 576 failures in 576 trials, and six GT-role diagnostic endpoints also total zero successes in 576. These are local diagnostic budgets, separate from the fourteen-task study.

A zero action-source deployment intervention changes the three 256-update scores from 0/12/26 to 0/1/46 successes out of 96 each. It was selected after observing existing outcomes and is exploratory. A reproducibility investigation attributes some same-seed trajectory branching to RGB channel changes of one gray level; fixed full-input/RNG inference and fixed-action physical replay remain consistent. This finding does not explain every historical trajectory difference.

G.2. Gripper Execution Diagnostics

The gripper investigation compares sign decoding, a fixed 0.1 dead zone, and nearest-label ternary projection and history-hold mappings. The ternary projection maps to $\{-1, 0, +1\}$ with thresholds at ± 0.5. For the reported fifth-epoch endpoint, the 1,260-scene counts are 698 for sign, 742 for the fixed dead zone, 749 for nearest-label ternary projection, and 741 for history hold. The ternary projection does not show a clear overall improvement over the fixed dead zone; history hold adds failures outside the capture subset. Neither is adopted as the production default in the evidence cited here.

The broader fourth-to-eighth-epoch dead-zone report contains descriptive counts but retains an unsuccessful strict historical trajectory-consistency audit. It is not used as an unrestricted main-table performance claim. These diagnostics show why a decoder change must be distinguished from a representation change and why both recovered and newly failed episodes need to be reported.

G.3. Figures and Aligned Rollouts

Main-paper Fig. 1 illustrates interaction geometry conceptually. Fig. 2 depicts the implemented D reference with P03's auxiliary targets. Fig. 3 plots measured success-rate differences and paired scene outcomes. The actual-scene task montage from the earlier benchmark assets can be used to illustrate the suite after its release identity is checked.

The remaining qualitative comparison aligns source observation, predicted future, actual future, and the contact event at the same physical times. It should include P05's adverse sphere condition, a retained sparse-trajectory condition, and a matched future-to-action intervention after that interface exists. Each case records method, scene, camera, frame, units, visibility, action application, and outcome. These assets are **[pending synchronized prediction/rollout exports]**.

Case selection must include the failures that alter the main conclusion. A successful clip illustrates an accepted outcome; it does not replace the full success denominator. Anonymous supplementary media should report filename, episode, duration, playback rate, cuts, resets, and human interventions. No real-robot video or forecast intervention video is claimed as delivered by the present manuscript.